

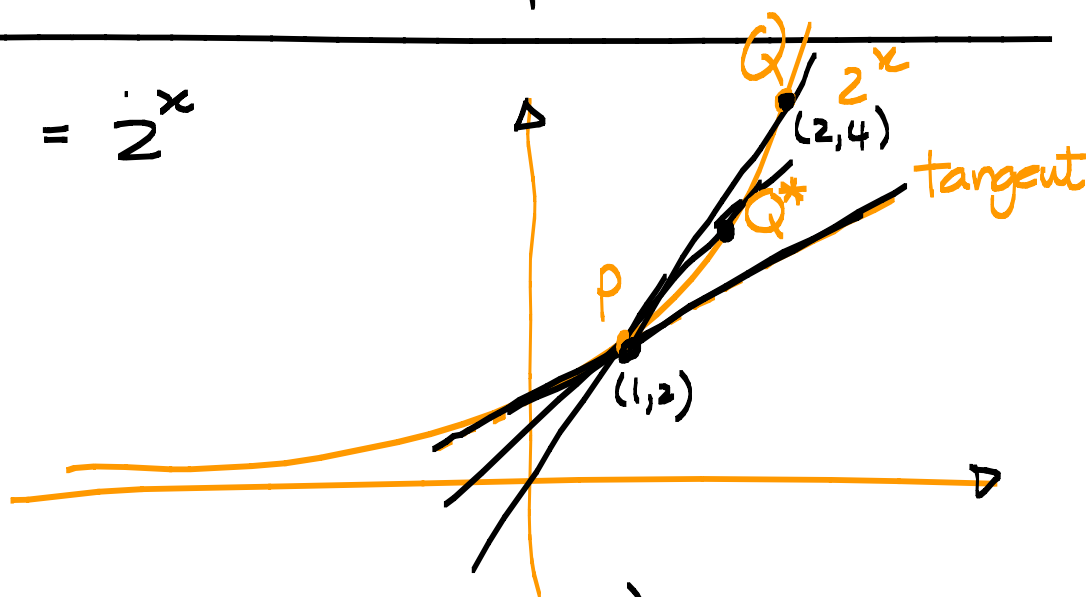
Theme 2: Limits, Continuity & Derivatives.

Note Title

2014/02/19

Unit 2.1 Tangent & Velocity problems. p 82 - 86

Sketch $y = 2^x$



Equation PQ (Secant)

$$y - y_1 = m(x - x_1)$$

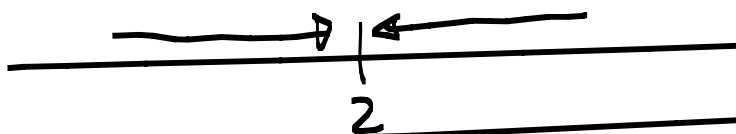
$$m = \frac{4-2}{2-1} = 2$$

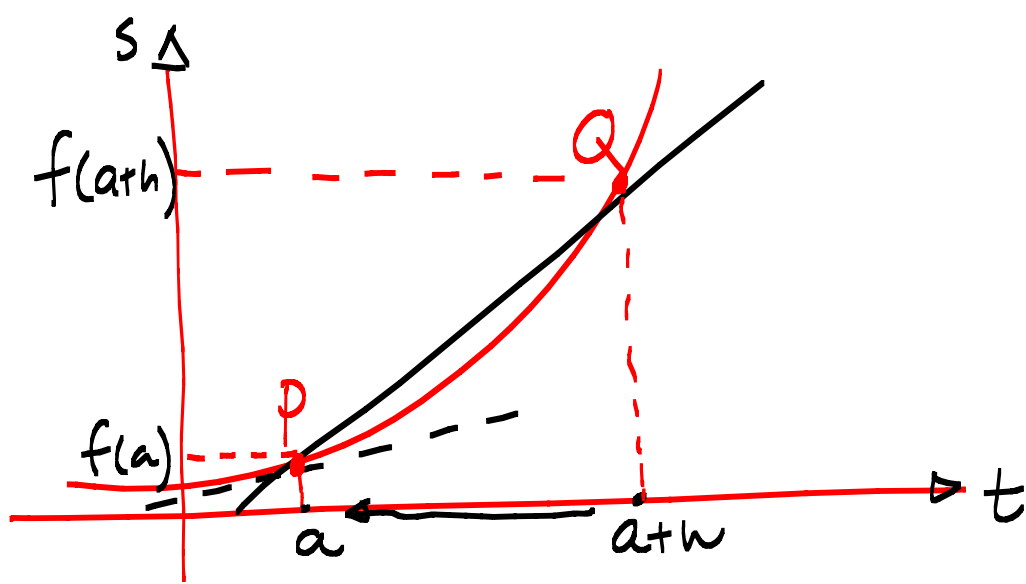
$$y = 2x$$

Equation PQ*

$$Q^* (1.7, 3.25)$$

$$y = 1.79x + 0.21$$





Avg rate of change - slope of the secant through PQ
 (Avg velocity)

$$m = \frac{f(a+h) - f(a)}{(a+h) - a}$$

Instantaneous rate of change in P
 (Inst. velocity)
 - slope of the tangent line.

We get this by calculating successively smaller intervals

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

limit of a function